SOFTWARE ENGINEERING — LAB 3

Component Modelling & Architectural Pattern Selection

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Architecture Selection

We chose **Layered Architecture** for the Self-Service Coffee Kiosk System.

Reasons for the Choice

1. Clear separation of responsibilities: The Presentation Layer handles touchscreen interactions, the Business Layer manages orders and credit-card payment requests, and the Data/Hardware Layer stores menu and pricing information and connects to the receipt printer. This structure makes the kiosk easier to understand, test and maintain.

2. Suitable for a single kiosk application: The system has a small, defined set of functions—selecting a coffee and size, paying by credit card, and printing a receipt. These functions can work together through clearly defined layers without the deployment and communication overhead of multiple microservices.

Security Advantage

The Payment Service can handle card-payment communication separately from the touchscreen interface. The interface should pass payment requests to this service rather than directly accessing sensitive payment information. Access to menu and pricing data can also be restricted to the components that need it.

Performance Benefit

The kiosk can process customer interactions within the application layers and retrieve menu and pricing data from its database component. This avoids unnecessary communication between independently deployed services and supports a quick ordering experience.

Conclusion

Layered Architecture is a practical choice for this kiosk because it separates user interaction, order processing, payment, and data or hardware access while keeping the overall design straightforward.